UNIVERSITY OF PERADENIYA
FACULTY OF SCIENCE

SEMESTER II - 2022/2023

MAT 1023 - REAL ANALYSIS I

MID SEMESTER EXAMINATION

Time allowed: Fifty (50) minutes **Answer ALL questions**

Instructions : This examination consists of two parts. Answer both parts as instructed.

Part A - Multiple Choice Questions (30 points) : There are six questions (5 points for each). For each question below, select the correct answer among the four choices and clearly color the respective item labeling.

1. Which of the following is not a proposition ?
   (a) $1 + 1 = 3$
   (b) Answer to this question.
   (c) Kamindu Mendis is a right-handed batsman.
   (d) Trump will be the next US President.

2. What is the negation of the statement "All birds can fly" ?
   (a) Some birds cannot fly.
   (b) No birds can fly.
   (c) All birds cannot fly.
   (d) Birds can sometimes fly.

3. The statement " $\forall x\ (P(x) \iff Q(x))$ " is equivalent to:
   (a) $\forall x\ (P(x) \implies Q(x)) \wedge \forall x\ (Q(x) \implies P(x))$
   (b) $\exists x\ (P(x) \implies Q(x)) \wedge \exists x\ (Q(x) \implies P(x))$
   (c) $\forall x\ (P(x) \implies Q(x)) \vee \forall x\ (Q(x) \implies P(x))$
   (d) $\exists x\ (P(x) \implies Q(x)) \vee \exists x\ (Q(x) \implies P(x))$

4. Which of the proposition $p \wedge (\neg p \vee q)$ is
   (a) Tautology
   (b) Contradiction
   (c) Logically equivalent to $p \wedge q$
   (d) All of the above

5. Which of the following propositions is a tautology ?
   (a) $(p \vee q) \implies q$ (b) $p \vee (q \implies p)$ (c) $p \vee (p \implies q)$ (d) both (b) and (c)

6. What is the truth value of the statement, " If 2024 is an odd number, then 7 is a prime number ".
   (a) TRUE
   (b) FALSE
   (c) Indeterminate
   (d) Contradiction

[Turn over...

Part B - Structured Questions (70 points) : Answer all three questions in the provided space. To receive full marks, ensure that each solution is thoroughly explained, with clearly justifying the steps of the chosen proof technique.

1. **(10 points)** Let $a \in \mathbb{Z}$. If $(a+1)^2 - 1$ is even, then prove that a is even.

2. **(30 points)** Let $P(n)$ be the statement that $1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{n^2} < 2 - \frac{1}{n}$, where n is an integer greater than 1.

   By completing each step below, use *mathematical induction* to prove that $P(n)$ is true for all integers n greater than 1.

   (a) Show that $P(2)$ is true, completing the basis step.

   (b) What is the inductive hypothesis ?

   (c) Complete the inductive step.

[Turn over...

3. **(30 points)** Let $\mathbb{Q} = \left\{ \frac{a}{b} \ : \ a, b \in \mathbb{Z}, b \neq 0 \right\}$ be the set of rational numbers and let $\mathbb{I} = \mathbb{R} \backslash \mathbb{Q}$ be the set of irrational numbers.

   (a) If $x, y \in \mathbb{Q}$, then prove that $x + y \in \mathbb{Q}$.

   (b) If $x, y \in \mathbb{Q}$, then prove that $xy \in \mathbb{Q}$.

   (c) Is it true that, if $x, y \in \mathbb{I}$, then $x + y \in \mathbb{I}$?

   (d) Is it true that, if $x, y \in \mathbb{I}$, then $xy \in \mathbb{I}$?